CLAIMS

What is claimed is:

1. An adaptive troposcatter communication system, comprising:

    - a first adaptive transceiver, comprising:
        - a channel condition monitor configured to assess real-time characteristics of a troposcatter communication channel;
        - a dynamic symbol-to-state mapper configured to, based on the assessed channel characteristics, dynamically define a set of N discrete, distinguishable radio frequency (RF) states, where N is an integer greater than two, and establish a mapping between abstract symbols and said N RF states;
        - a high-radix RF modulator configured to receive a sequence of said abstract symbols and generate a corresponding sequence of said N RF states; and
    - a second adaptive transceiver, communicatively coupled to the first transceiver via the troposcatter communication channel, the second adaptive transceiver comprising a high-radix RF demodulator configured to, based on an inverse of the mapping, decode a received sequence of said N RF states into a sequence of abstract symbols.

2. The system of claim 1, wherein the dynamic symbol-to-state mapper is configured to adjust the value of N based on the channel conditions.

3. The system of claim 2, wherein the mapper is configured to reduce N during periods of channel degradation to enhance reliability.

4. The system of claim 2, wherein the mapper is configured to increase N during periods of favorable channel conditions to enhance data throughput.

5. The system of claim 1, wherein the abstract symbols are components of a parametric data packet, the payload of which is an unevaluated mathematical expression.

6. A method for adaptive troposcatter communication, comprising:

    - continuously assessing real-time characteristics of a troposcatter communication channel;
    - dynamically defining a set of N discrete, distinguishable radio frequency (RF) states based on said assessed channel characteristics, where N is an integer greater than two;
    - establishing a mapping between abstract symbols and said N RF states;
    - encoding a sequence of said abstract symbols into a corresponding sequence of said N RF states;
    - transmitting said sequence of RF states via a troposcatter link; and
    - receiving and decoding said sequence of RF states back into the original sequence of abstract symbols using an inverse of the mapping.

7. The method of claim 6, wherein the value of N is dynamically adjusted based on the assessed channel characteristics.

8. The method of claim 6, wherein the abstract symbols represent an unevaluated mathematical expression that is transported across the troposcatter link without loss of precision.